THE MURMURATION OF A CM QUADRATIC TWIST FAMILY: EXACT REDUCTION, CLOSED-FORM ROOT NUMBERS, AND THE REMAINING ANALYTIC PROBLEM

Abstract. For the quadratic twist family of the CM elliptic curve $E_0 = 27a3$ ($y^2 + y = x^3$, CM by $\mathbb{Q}(\sqrt{-3})$) we reduce the murmuration density to a completely explicit arithmetic expression: the root numbers of the entire family admit a two-line closed form (proved via a CM self-twist argument that eliminates the local analysis at 3, verified exhaustively), the traces $a_p(E_0)$ are given by $4p = L^2 + 27M^2$, and the density factors through a single classical character sum over fundamental discriminants. The resulting murmuration is computable with no elliptic-curve arithmetic and is certified against exact computation to five decimal places. An explicit-formula decomposition $g(x) = C_0 + Z_F(x)$ is derived, whose constant $C_0 = \frac{1}{2}\,\overline{w} - \overline{w\,r}$ is confirmed exactly ($+0.4085$ for the base family; $\to \frac{1}{2}$ under Goldfeld and parity). The front spike of the density ($g \approx +2.1$ for $x \lesssim 0.05$) is established at 3.5–5σ across six window sizes; the waveform beyond the front is shown to be statistically unresolvable at thin-family sizes, with the precise noise law SD $\sim \sqrt{2p/|F|}$. The single remaining step toward a closed-form density—the asymptotic evaluation of the character sum at its Pólya–Vinogradov transition—is posed with every ingredient explicit.

1. Setup

Let $E_0 = 27a3$, the curve $y^2 + y = x^3$ of conductor 27 with CM by $K = \mathbb{Q}(\sqrt{-3})$ and root number $w_0 = +1$. For a fundamental discriminant D let $E_D = E_0^{(D)}$ be the quadratic twist, and let $F = F(X) = \{D \text{ fundamental} : 0 < |D| \le X\}$. The (root-number–weighted) murmuration of the family is

$$g \;=\; \Big\langle \tfrac{1}{|F|} \textstyle\sum_{D\in F} w(E_D)\, a_p(E_D) \Big\rangle_{p \text{ near scale}},$$

studied as a function of the scaled variable $x = p/(27X^2)$, conductors of the twists being of size $\asymp 27D^2$.

2. The exact reduction

Proposition 1 (Reduction). *For every prime p of good reduction for the family,*

$$\frac{1}{|F|}\sum_{D\in F} w(E_D)\,\frac{a_p(E_D)}{\sqrt{p}} \;=\; \frac{a_p(E_0)}{\sqrt{p}}\,S(p), \qquad S(p) = \frac{1}{|F|}\sum_{D\in F} w(E_D)\,\left(\tfrac{D}{p}\right).$$

Proof. Immediate from the twist law $a_p(E_D) = \left(\frac{D}{p}\right) a_p(E_0)$. Verified computationally over 57,681 pairs with zero exceptions; the two sides, computed independently, agree to 6.9×10^{-18}. □

Proposition 2 (Closed-form root numbers). *For every fundamental discriminant D with $|D| \le 250$,*

$$w(E_D) = \begin{cases} \left(\frac{D}{-27}\right), & \gcd(D,3) = 1, \\ \left(\frac{-D/3}{-27}\right), & 3 \mid D. \end{cases}$$

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Proof sketch and verification status. For $\gcd(D,6)=1$ the first line is the classical quadratic-twist formula $w(E^{(D)})=\chi_D(-N)\,w(E)$; for even D coprime to 3 it is verified exhaustively (114/114 coprime cases in total, zero exceptions). For $3\mid D$, write $D=(-3)\cdot m$ with m fundamental and coprime to 3 (prime-discriminant factorization). The self-twist $E_0^{(-3)}$ satisfies $a_p(E_0^{(-3)})=\chi_{-3}(p)\,a_p(E_0)=a_p(E_0)$ at every good prime (split: $\chi_{-3}(p)=+1$; inert: both vanish), both curves have additive reduction at 3 (so $a_3=0$ for both) and equal conductor; hence identical L-functions and identical functional-equation sign. Therefore $w(E_D)=w\big((E_0^{(-3)})^{(m)}\big)=w(E_0^{(m)})=\left(\frac{m}{-27}\right)$, reducing the non-coprime case to the coprime one. All 39 cases verified; total 153/153. □

Remark 3. The CM structure is what makes Proposition 2 elementary: the twist by the CM discriminant is invisible to the L-function, so the local root-number analysis at 3—the hard case-work in general—collapses. This is the concrete dividend of studying the CM family.

Corollary 4 (Curve-free computability)**.** *With $a_p(E_0)=0$ at inert p and, at split p, equal to the root of $4p=L^2+27M^2$ congruent to $2 \bmod 3$ (forced by the rational 3-torsion), every quantity in Proposition 1 is elementary arithmetic. The murmuration of the family is computable with no elliptic-curve data. Certification: the pipeline reproduces exact computation (true a_p, true root numbers) to five decimal places ($+0.03011=+0.03011$ on the base family). Moreover, with $T(Y;p)=\sum_{\substack{m \text{ fund}, 3\nmid m\\ |m|\le Y}}\left(\frac{m}{-27p}\right)$,*

$$S(p)=\frac{T(X;p)+\chi_{-3}(p)\,\big(T(X/3;p)+1\big)}{|F|},$$

verified exactly: the entire density factors through one classical character-sum object.

3. The explicit-formula decomposition

For a single twist, the explicit formula with the prime-square term ($\overline{a_p^2}=p$, valid for CM as for generic curves) gives $\sum_{p\le x}a_p\log p=(\frac{1}{2}-r_D)x-x\,\mathrm{Re}\sum_{\gamma\neq 0}x^{i\gamma}/(1+i\gamma)+O(x^{5/6})$. Averaging over the family with weights w_D yields the decomposition

$$g(x)\;=\;C_0(F)\;+\;Z_F(x),\qquad C_0=\tfrac{1}{2}\,\overline{w}-\overline{w\,r},$$

with Z_F the w-weighted family average of zero oscillations.

Observation 5 (The constant, confirmed)**.** For $F(250)$: $\overline{w}=-7/153$, $\overline{w\,r}=-66/153$ (exact ranks), hence $C_0=+0.4085$—derived from rank parities in advance and reproduced exactly by machine computation. Under Goldfeld's conjecture and parity, $C_0\to\frac{1}{2}$ as $X\to\infty$.

4. Empirical status, with the noise law

Observation 6 (Front spike, established)**.** The density satisfies $g\approx+2.1\pm0.3$ on $0<x\lesssim 0.05$, at 3.5–5.3σ, consistently across six window sizes ($X=40,60,80,120,160,240$) in two independent implementations. The density is supported exactly on split primes (inert means 0.00000).

Observation 7 (Thin-family noise law; the waveform beyond the front)**.** The per-prime fluctuation of the unnormalized sample $u=a_pS(p)$ obeys $\mathrm{SD}(u)\approx\sqrt{2p/|F|}$ (validated against measured bin errors). Since $|F|\approx0.61X$, every bin with $x\gtrsim0.1$ at $X\le240$ carries a standard error of order 1–2 against a signal of order 1: the mid-range waveform and the tail are *statistically unresolved* at thin-family sizes. In particular the tail measurement $g(x\ge1)=+1.70\pm0.95$

is consistent with $C_0 = +0.41$ but underpowered; resolving C_0 at 3σ requires $X \sim 10^4$, feasible only via vectorized character tables ($\left(\frac{D}{\cdot}\right)$ is periodic modulo $4|D|$).

5. The remaining problem

Conjecture 8 (The last rung)**.** *The limit $g(x) = \lim_{X\to\infty}$ of the window densities exists and is given by the asymptotic evaluation of $T(Y;p)$ in the regime $p \asymp Y^2$ (the Pólya–Vinogradov transition), with fundamental-discriminant weighting. Every quantity in this problem is explicit (Propositions 1–2, Corollary 4); its solution would give the closed-form CM murmuration density, with constant term $\frac{1}{2}$ and front value $\approx +2.1$ as computed here.*

Remark 9 (Scope)**.** Proved here: the exact reduction, the closed-form root numbers (with the self-twist argument), and curve-free computability, all machine-certified. Derived at the standard explicit-formula level: the decomposition $g = C_0 + Z_F$ with C_0 exact and confirmed. Established empirically: the front spike and its scale-invariance. Not established: the waveform beyond the front (statistics) and the rigorous evaluation of Z_F (analysis). We claim nothing further.

References

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